

RESEARCH REPORT 

Tab Manager and Mega Menu Concept Test



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BACKGROUND

Overview & Methods

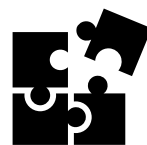
The context

Users need help managing tab clutter. The solutions we develop should meet these design principles:



A tool for everyone

Optimize for daily use, enable a transitioning between horizontal and vertical tabs



Bring similar concepts together

Combine similar views (vertical tabs, tab manager) and tools like Recently closed, workspaces



Simplify the states

Minimizing the number of different states or modes the user interface can be in

The Research

Research goal

Determine if tab manager and mega-menu meet users’ need to manage tab clutter while aligning with our design principles

Method

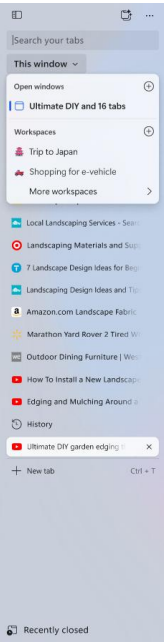
16 remote unmoderated sessions on UserTesting

Participants

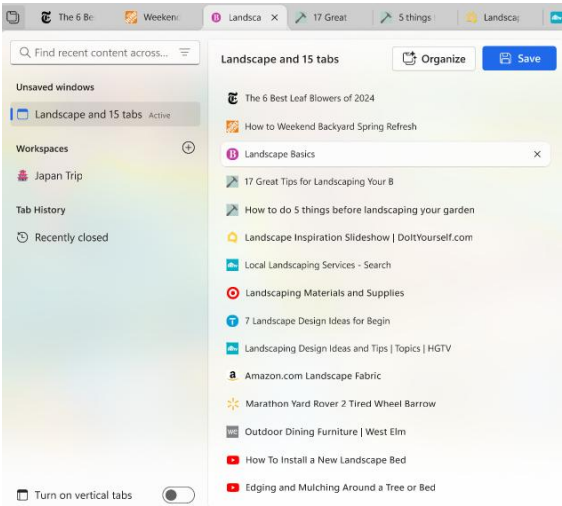
Workspace & tab group users, non-workspaces & non-tab group users

Materials

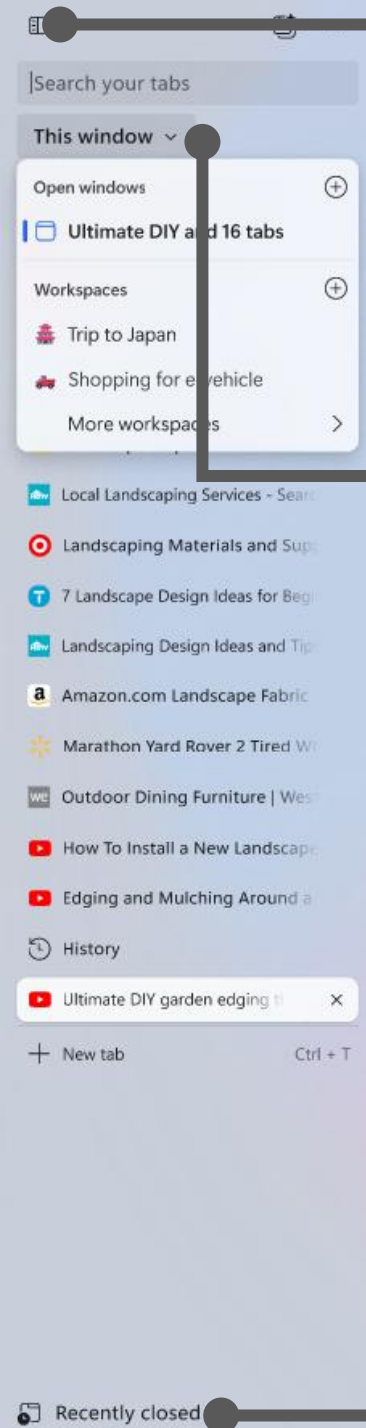
Two clickable prototypes in figma:



Tab manager



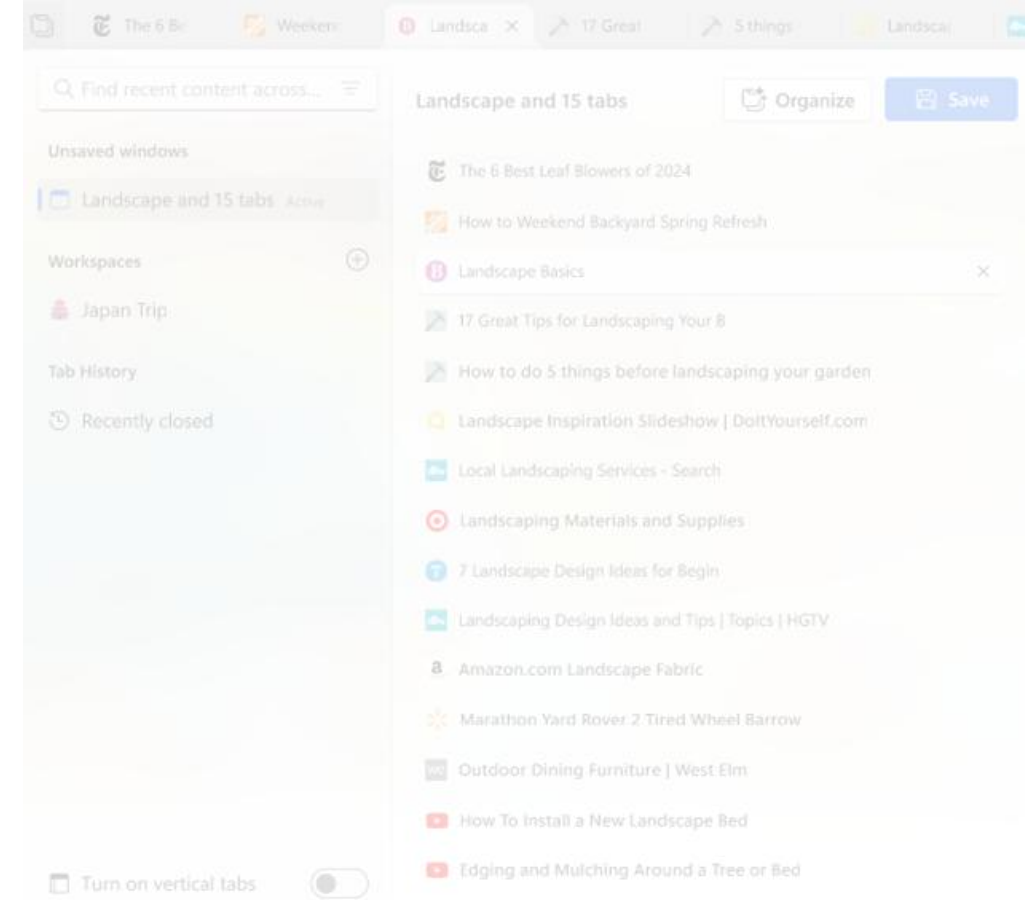
Mega menu

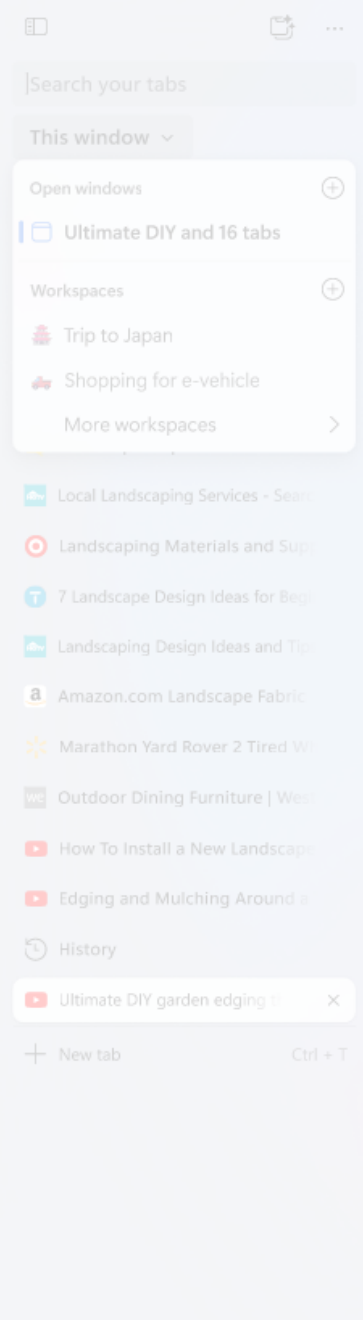


Tab manager icon
Entry point for tab manager

This window
Workspaces in the dropdown

Recently closed
Available on opening tab manager

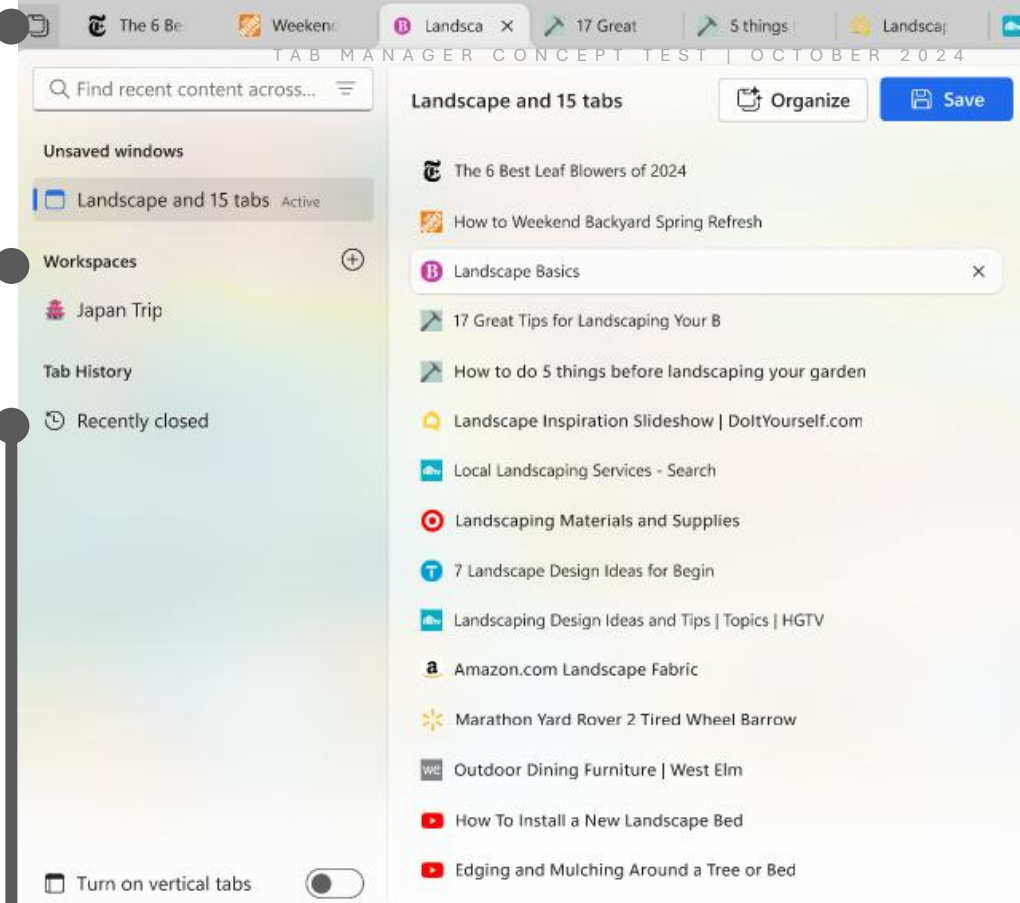




Mega menu icon
Entry point for tab manager

Workspaces
Available on opening mega menu

Tab history
Available on opening mega menu



The scope

This round of user research focused on how tab manager and mega-menu align with these design principles:



A tool for everyone

Optimize for daily use, enable a transitioning between horizontal and vertical tabs



Bring similar concepts together

Combine similar views (vertical tabs, tab manager) and tools like Recently closed, workspaces



Simplify the states

Minimizing the number of different states or modes the user interface can be in

Research questions

Here our research questions map onto design principles:



A tool for everyone

When and why do different user groups need a tab tool

How will tab manager and mega-menu help users quickly find a tab

How do these tools facilitate the transition between horizontal and vertical tabs

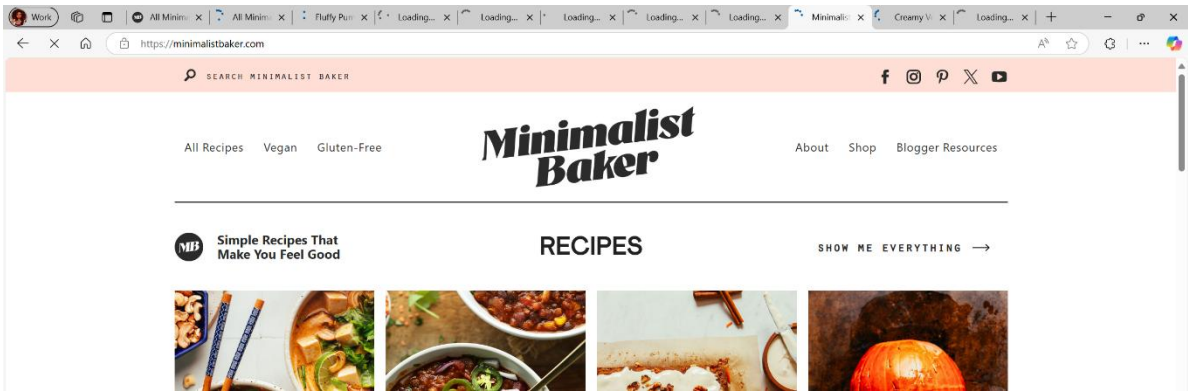


Bring similar concepts together

Will users have easier access to features like Recently closed tabs and Workspaces from these entry points

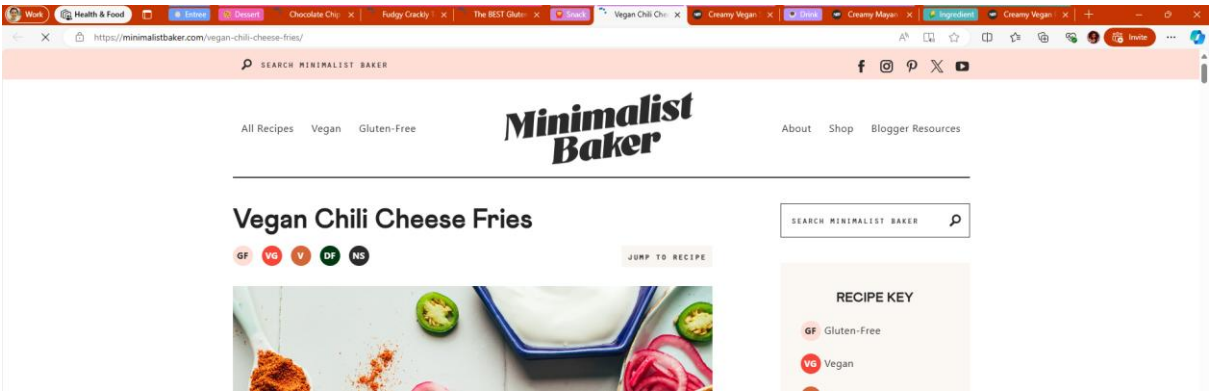
Participants

We tested the concepts with two user groups in remote unmoderated sessions. Both have **10+ tabs open** at a time and **have not used vertical tabs** in the past month.



10

Non-workspaces
& non-tab group users¹



6

Workspaces
& tab groups users

¹Also have not used tab search in the past month

Results

Tab Manager



A tool for everyone

When and why do different user groups need a tab tool

When and why do different user groups need a tab tool

In the current state, how horizontal tab users are accomplishing these tasks:

Non-workspaces & non-tab group users

Workspaces & tab groups users

Managing
tab clutter

- Favorites
- Close some tabs off
- Organizing from left to right

- Favorites
- Workspaces
- Tab grouping

Tab search

- Hover preview
- Memorizing the tabs
- Alt tab
- Opening each tab

- Hover preview
- Memorizing the tabs
- Tab search
- Tab grouping

Revisiting a
closed tab

- History
- Favorites

- History
- Favorites

**Users are saving their
tabs externally**

**Non-workspace
& non-tab groups user:**

“when I'm really busy and I've got a lot of tabs open, the length of the tab title shortens and I can no longer see, I gather them up and put them into a document, take a break and start again” –P8

**Users are not managing
their tabs**

**Non-workspace
& non-tab groups user:**

“*I don't do anything, I'll
have tabs upon tabs, will
be hopping between them*”
–P15



A tool for everyone

Can tab manager help users quickly find a tab

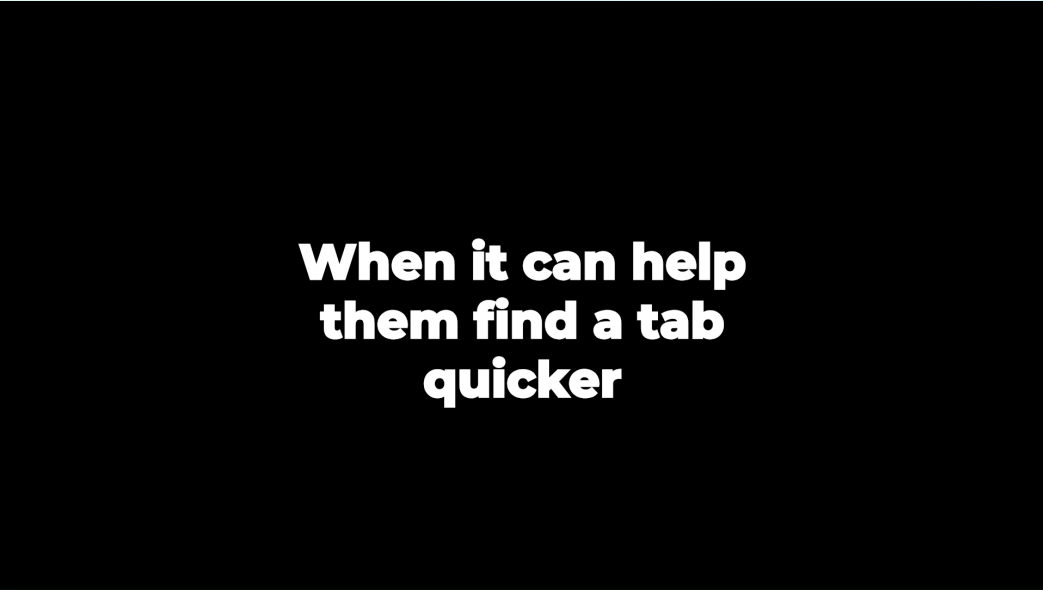
Quickly finding a tab

In their own words, how tab manager helps users quickly find a tab:

Positive

“I can use search”

“It expands the name”

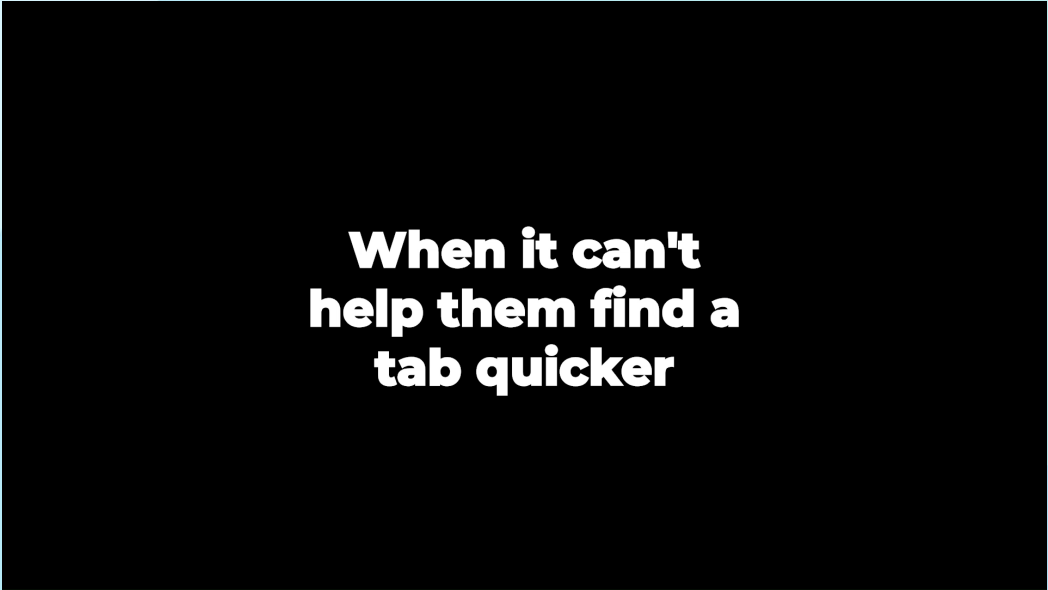


Positive

Negative

“Without search it might not be quicker”

“eyes drawn to more things”



Negative

Quickly finding a tab

Both user groups called out ways the tab manager could help them find a tab more quickly, and ways it can't:



I can see the whole title

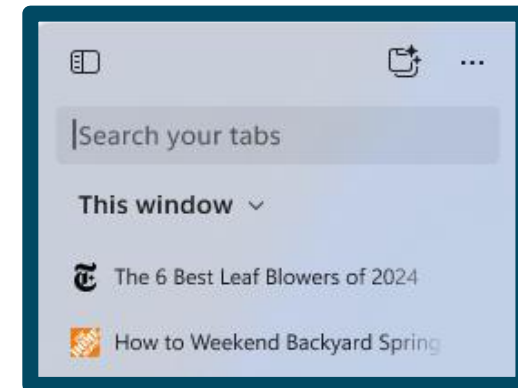
6 participants talked about being able to see more of the tab names

“When you have so many open you cannot read besides seeing the logo, this will help you, you can read most of it” –P1



Search

3 participants talked about how the search function could help them find a tab



When there are few tabs

4 participants called out that they wouldn't need the tool if their tabs were easy to find



Without search it's not helpful

1 participant felt that without search and in a scenario where there were 80 tabs, it wouldn't be easy to find a tab



Too much to look at

1 participant felt there was too much going on

“Your eyes are drawn to more things, it's a double edged sword, you might not be able to see it as quickly if it's right in front of you.” – P4

Recommendation | More testing in code to determine how tab manager can help in situations where there are few open tabs vs. many open tabs

**Finding a tab comes
at the cost of screen
real estate**

**Non-workspace
& non-tab groups user:**

“*It consumes vertical
space on the display
which compromises my
user experience”*



A tool for everyone

Can tab manager help users quickly find a tab



**Horizontal tab users can
find a tab more quickly**



A tool for everyone

Does tab manager facilitate a transition to vertical tabs

**Tab manager is
similar to vertical
tabs now**

**Non-workspace
& non-tab groups user:**

“ *Actually, for me I think
nothing is different.
It's just like the vertical
tabs one, it's just the
same. I don't know.”*

Vertical tabs

In their own words, what was it like to flip to vertical tabs, and how tab manager impacts their usage of vertical tabs:

Reactions

“It’s right there”

“It looks more organized”

“This is a game changer”

“A bit worried”

“Takes up real estate”

How tab manager affects vertical tabs

“I didn’t know about it before”

**How familiar are you
with vertical tabs**

Vertical tabs

There's an awareness problem more than a desirability problem

Horizontal tab users may not know about vertical tabs.



Never used it

11 of the 16 participants had never turned it on. Of these, 5 had never heard of it before.

Participants identified ways they would use vertical tabs:

“for me as a student and employee who works remote on a computer vertical is perfect.” – P12

I feel like this could help me navigate towards my work streams a little easier. For example, JIRA messaging tickets, specific tickets I could pull back rather than going through my groupings –P3

“it is nice to see more of the name of the page, for example when you're reading an article with a long name” – P6

Insight | There is an awareness problem with vertical tabs, not necessarily a desirability problem.

Tab manager **may not be the catalyst** we're hoping it is

For many participants, the draw to vertical tabs has nothing to do with tab manager



Tab manager has nothing to do with it

“I don’t think [using vertical tabs] would change with tab manager, I’ll switch to using them all the time.” – P8



People who don’t like vertical tabs will not be convinced by tab manager

“No, vertical tabs takes up space” – P16

“I don't know, I just like the horizontal because I guess it's what I'm used to more than anything. It probably gives you more room than up top, but my preference is for the traditional horizontal across the top tabs... maybe if I had the option to do both I might use that.” –P4

Insight | People are drawn to vertical tabs without the help of tab manager.

It helps to see horizontal and vertical tabs at the same time, but also **creates clutter**

Participants talked about whether it could help them to have vertical and horizontal tabs at the same time:



Having both helps me adjust

4 participants talked about the familiarity of seeing the horizontal tabs there too

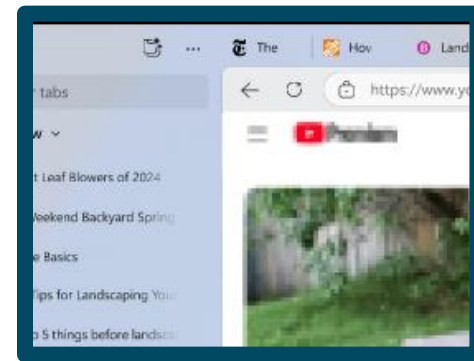
“It gives me a step between something new and something familiar” –P8



Too much on my screen

5 participants described the simultaneous tabs as confusing, redundant, and adding clutter

“Honestly it doesn’t help me... tabs at the top are more cumbersome and difficult to find.” – P14



Insight | Though it may ease the transition, simultaneous horizontal and vertical tabs violates a key design principle to reduce clutter.



A tool for everyone

Does tab manager facilitate a transition to vertical tabs

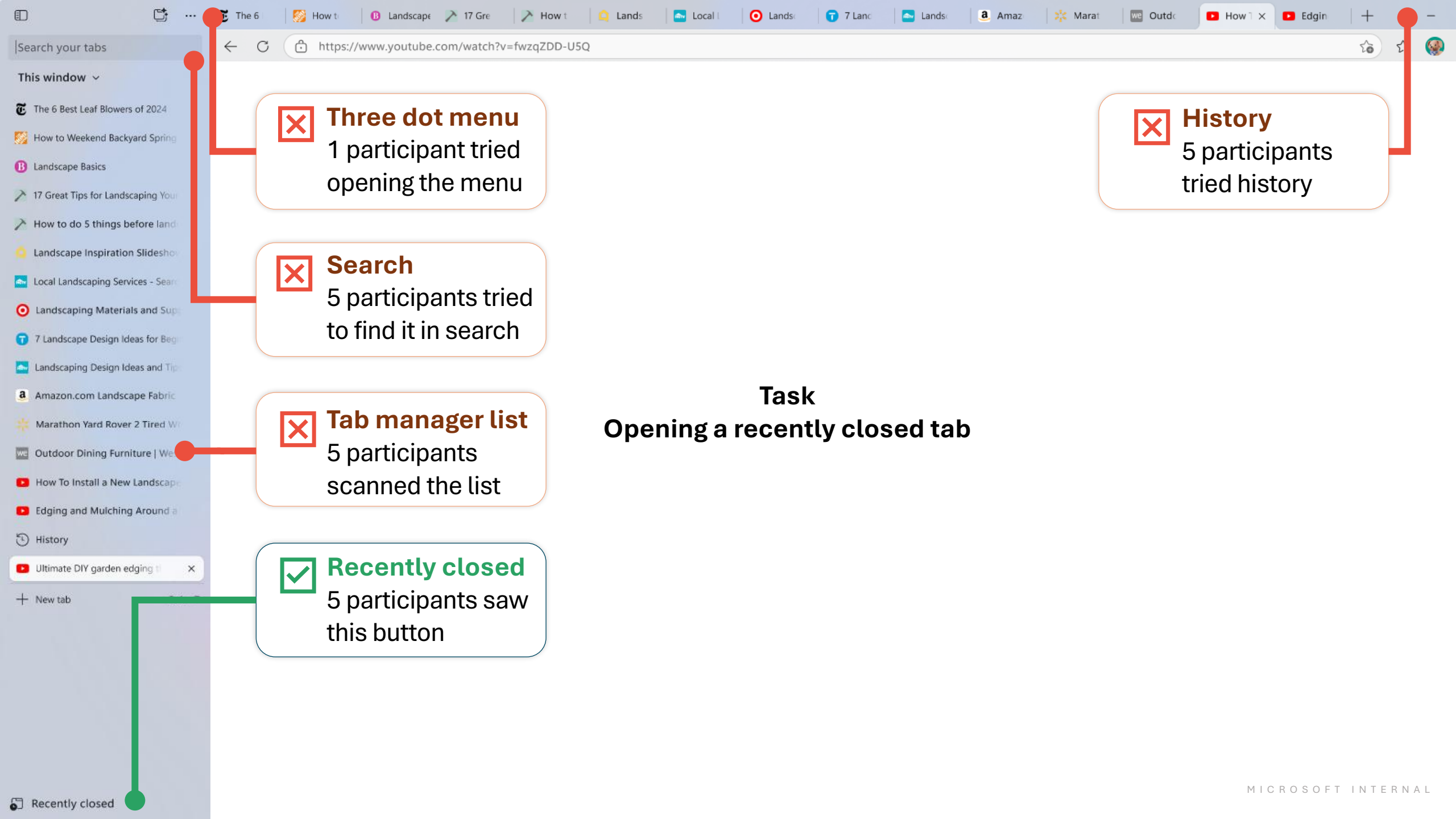


**There is an awareness
problem with vertical tabs**



Bring similar concepts together

Can tab manager help them revisit a closed tab



✗ **Three dot menu**
1 participant tried
opening the menu

✗ **Search**
5 participants tried
to find it in search

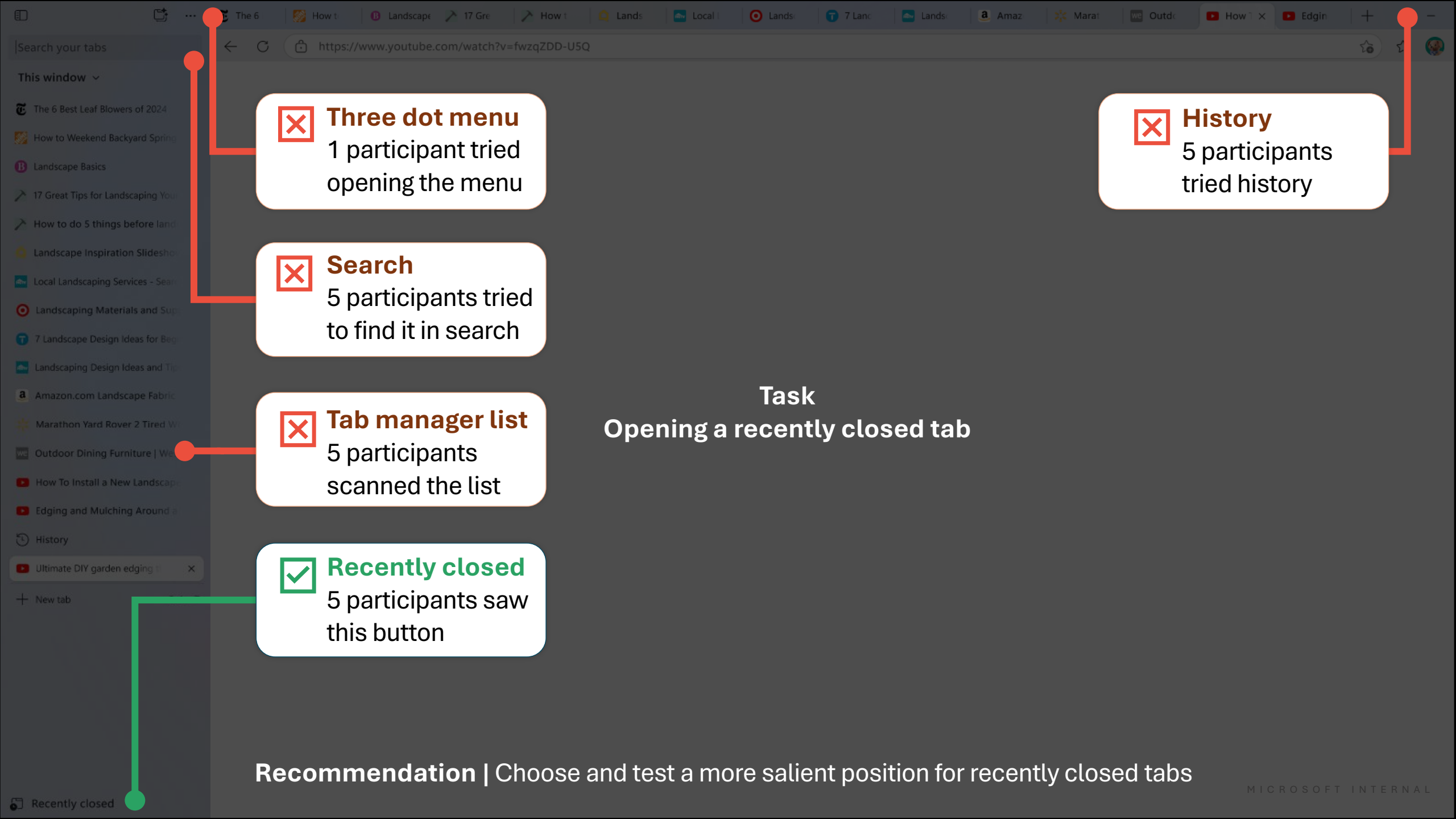
✗ **Tab manager list**
5 participants
scanned the list

✓ **Recently closed**
5 participants saw
this button

✗ **History**
5 participants
tried history

Task

Opening a recently closed tab



Three dot menu

1 participant tried opening the menu



History

5 participants tried history



Search

5 participants tried to find it in search



Tab manager list

5 participants scanned the list



Recently closed

5 participants saw this button

Task
Opening a recently closed tab

Recommendation | Choose and test a more salient position for recently closed tabs

Revisiting a closed tab

In their own words, what was it like to revisit a closed tab:

Positive

“It’s fewer clicks”

“Everything is in one place”

**What's more
efficient about
reopening a
closed tab**

Positive

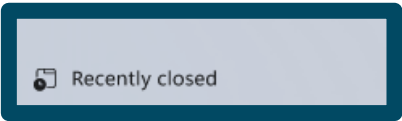
Negative

“It’s the same amount of clicks”

“Ctrl-shift-t is good enough for me”

**Nothings more
efficient about
reopening a
closed tab**

Negative



Revisiting a closed tab

Compared to what they do right now, here's how tab manager helps and doesn't help with revisiting a closed tab:



It makes use of tools I already use

1 workspace and tab group user felt that it was a good natural progression from what they're already doing

“It feels like a natural progression, efficient and easy to use based on how I already use it.” –P5



Equal or more clicks

7 participants said their method requires fewer clicks (e.g., Ctrl-Shift-T) or the same number of clicks (e.g., tab History).



Hard to find

3 participants felt what was inefficient was the findability of the button

Insight & Recommendation | In most scenarios revisiting a closed tab with tab manager is not easier than using existing tools. Test in code when and why users need something heftier than their usual tools.



Bring similar concepts together

Revisiting a Closed Tab

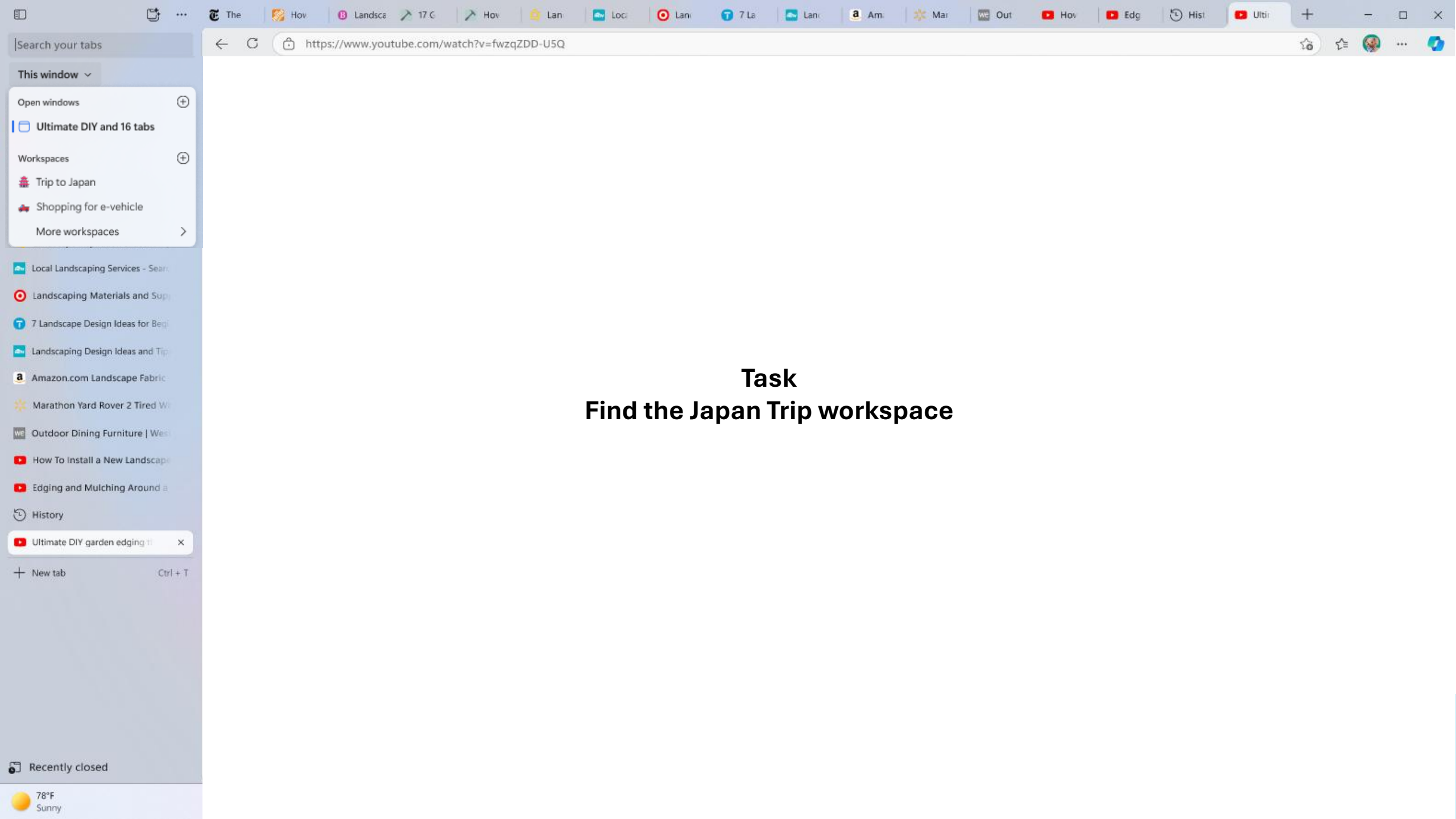


**Existing processes
are more efficient**

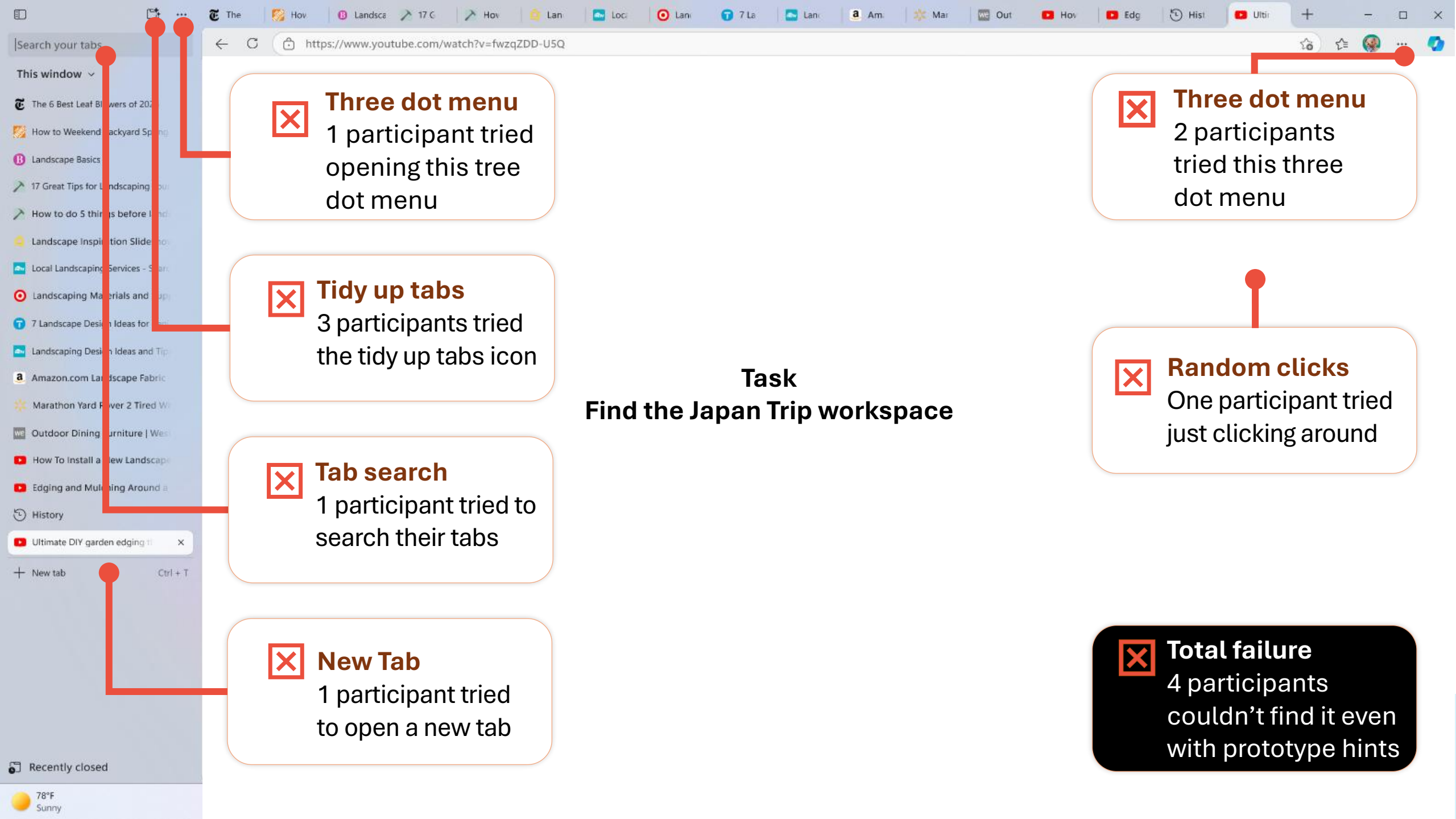


Bring similar concepts together

Workspaces



Task
Find the Japan Trip workspace



Three dot menu

1 participant tried opening this tree dot menu



Tidy up tabs

3 participants tried the tidy up tabs icon



Tab search

1 participant tried to search their tabs



New Tab

1 participant tried to open a new tab



Three dot menu

2 participants tried this three dot menu



Random clicks

One participant tried just clicking around

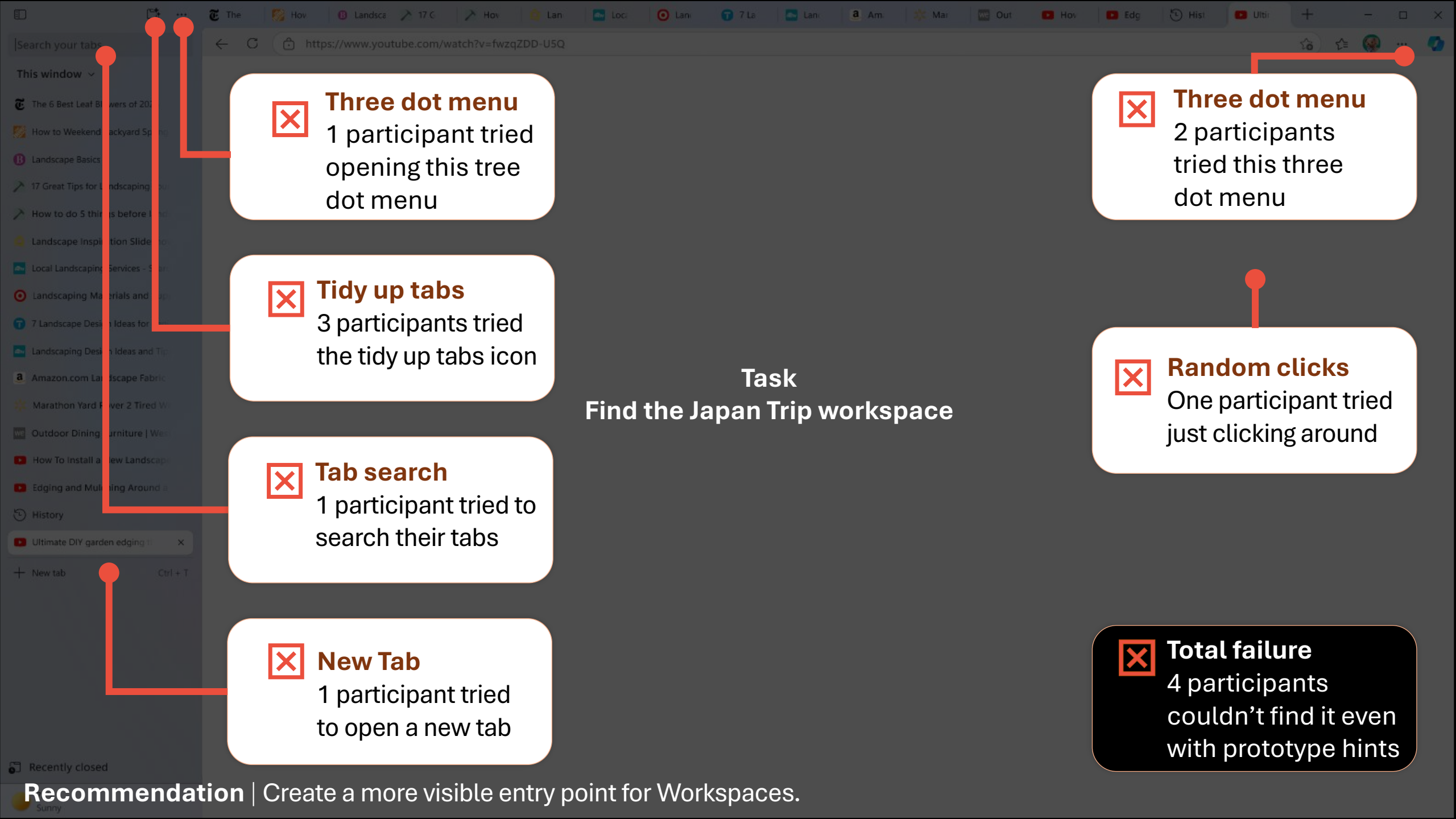


Total failure

4 participants couldn't find it even with prototype hints

Task

Find the Japan Trip workspace



Three dot menu

1 participant tried opening this tree dot menu



Tidy up tabs

3 participants tried the tidy up tabs icon



Tab search

1 participant tried to search their tabs



New Tab

1 participant tried to open a new tab

Task

Find the Japan Trip workspace



Three dot menu

2 participants tried this three dot menu



Random clicks

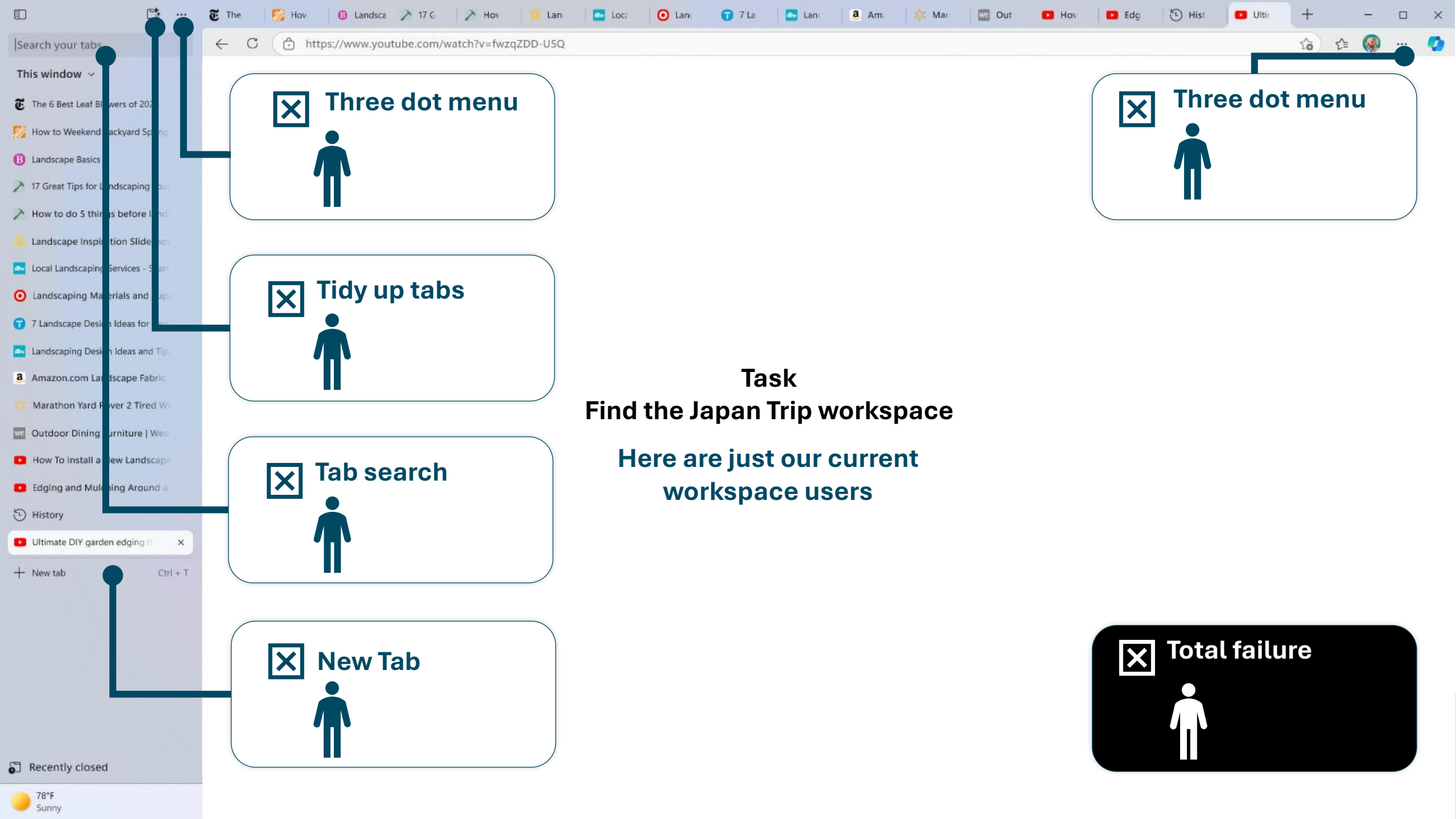
One participant tried just clicking around



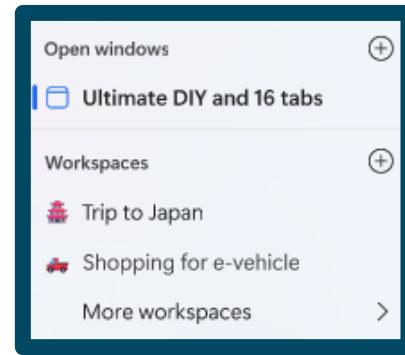
Total failure

4 participants couldn't find it even with prototype hints

Recommendation | Create a more visible entry point for Workspaces.



Current workspace users talked about ways tab manager made workspaces easier or more difficult



More seamless experience

3 participants talked about how it's natural or intuitive to have this function paired with other tab manager functions



Hard to find

3 participants observed that it's much harder to find now

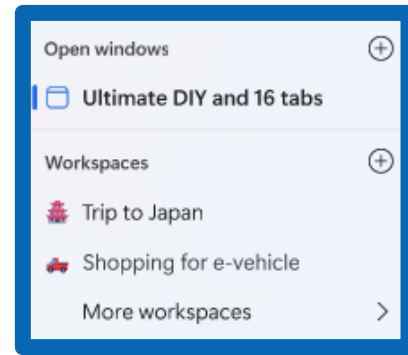


Burying my tabs somewhere else

1 participant felt that this could make it harder to search over multiple tabs across multiple workspaces

Insight | If we can solve discoverability power users may enjoy a more seamless experience.

Workspace non-users talked about whether tab manager would get them started with workspaces



It's that I didn't know about it

4 participants mentioned they might use it now that they're aware of this feature, not necessarily that tab manager would help them do it



Not for me

2 participants felt that this feature was for business use only and wouldn't solve any of their needs



This is inefficient

1 participant felt this buried tabs in another part of their browser, when they already had trouble with visibility

"I have to get to a whole other group and search again, essentially I'm doing two searches now" – P13

Insight | Non-workspace users may not want this feature given that they don't use it already.



Bring similar concepts together

Workspaces



**Not discoverable, won't
entice non-workspace users**

Results

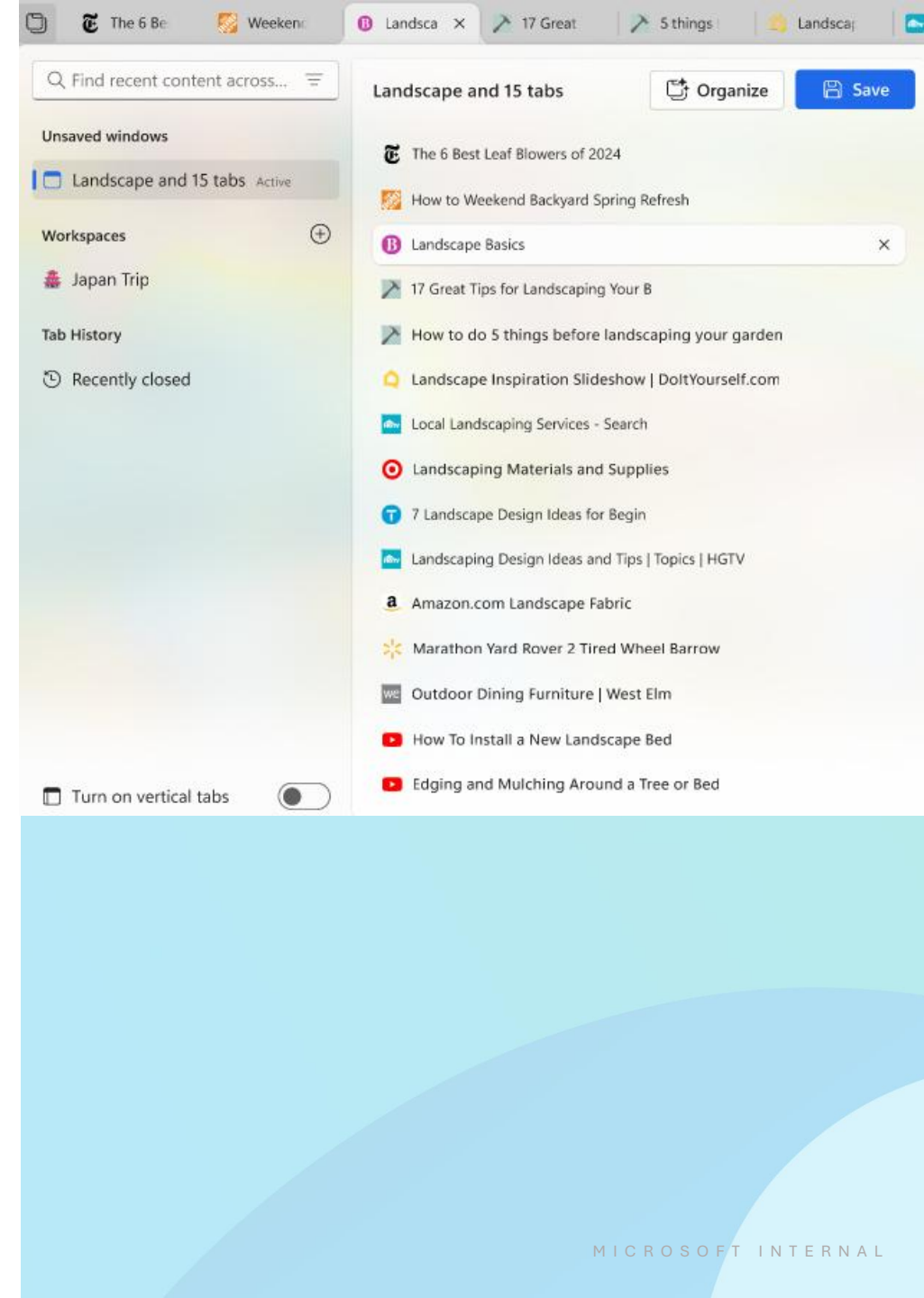
Mega Menu



A tool for everyone
& Bring similar concepts together

How well does mega menu align with our principles

In this round of testing,
the goal was to uncover
reasons **why not to
build mega menu in
code** (i.e., fast fail).



Mega menu

In their own words, what’s better about mega menu vs. tab manager:

Mega menu wins

“Better for nipping in and out” *“Reopen a closed tab is right there”*

**Which tab
manager can
help you find a
tab more quickly**

Mega menu wins

Tab manager wins

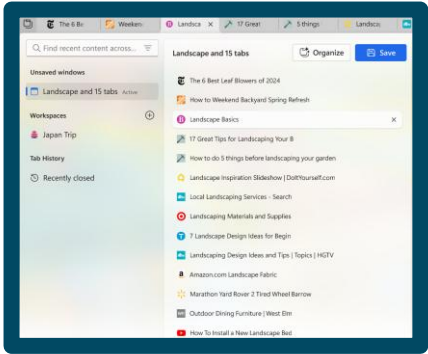
“I’m going to lose my mind if there’s a scroll bar” *“Opening a closed tab is more clicks”*

**Which tab
manager can
help you find a
tab more quickly**

Tab manager wins

Quickly finding a tab

Participants considered whether mega-menu would help them find a tab more quickly than tab manager:



 Quick to open and close

2 participants liked the efficiency of opening and closing mega menu whenever

“This one is better for nipping in and out quickly.” –P8

 It closes by default

2 participants felt mega-menu couldn’t help them find a tab as quickly because it doesn’t stay pinned

 It’s overwhelming

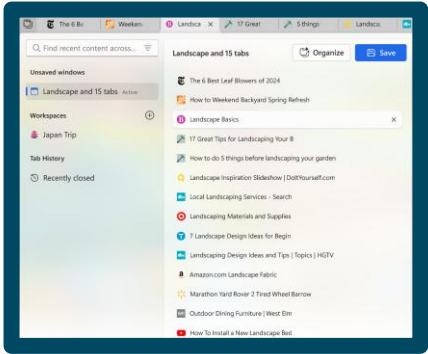
1 participant was concerned about the size of mega-menu

“This is insane, I’m going to lose my mind if we open 2 more tabs, if there’s a scroll bar here it’s going to be way too many.” – P13

Insight | Mega menu lets users nip in and out, tab manager helps users find a tab by staying pinned

Revisiting a closed tab

Here’s how mega menu compared to tab manager on bring similar concepts together:



The button is easier to find

5 participants observed that the reopen closed tabs button was more visible in mega menu.

“this one because it's directly in front of you, which I couldn't see at the beginning of the test.” –P9



Mega menu requires more clicks

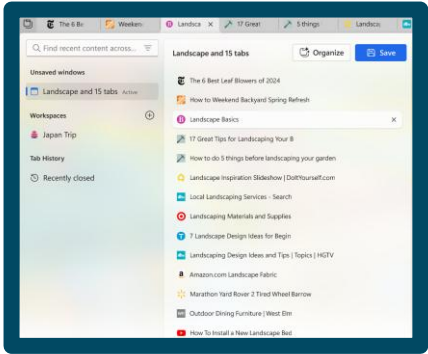
7 participants preferred tab manager, and of the 2 who specified why, it was because tab manager is pinned open and requires fewer clicks.

“I’s just kind of one more step to find... If I wanted to find a recently closed tab it would be in the bottom left corner all the time in the other tab manager” – P13

Insight | If tab manager is the same number of clicks as their usual method, mega menu adds an unnecessary click.

Workspaces

Here’s how mega menu compared to tab manager on bring similar concepts together:



 The entry point is visible

2 participants talked about how much easier it was to find workspaces in mega-menu than in tab manager

 They get a preview

3 participants noticed mega-menu let them flip between workspaces and see a complete list without clicking into a different one

 It’s not pinned open, so it takes more clicks

4 participants talked about the extra step it takes to open a workspace in mega-menu

 The information feels incomplete

1 participant thought mega-menu showed a summary or overview rather than a fully functioning workspace

Recommendation | In tab manager increase the visibility of workspaces, identify ways to help users glimpse a workspace before they enter one.

Design principles

Adding back to the mix, which of these concepts can simplify the states



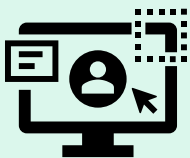
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Simplify the states

Minimizing the number of different states or modes the user interface can be in



A tool for everyone
& Bring similar concepts together

How well does mega menu align with our principles



**Requires
another click**



**Doesn't
simplify states**

Results

Key Takeaways

Key Takeaways

Tab manager

Mega menu



Quickly finding a tab



Tab manager helps horizontal tab users find a tab more quickly



Mega-menu doesn't stay pinned open, so it's not as quick to help users find a tab



Transitioning to vertical tabs



There is a discoverability problem with vertical tabs, not a desirability problem



Doesn't meet the design principle of minimizing the number of different states



Revisiting a closed tab & workspaces



Discoverability of the entry point is a problem, not more efficient than revisitation now, doesn't convert non-workspace users to workspace users



These features require a further click, where both concepts are already more cumbersome than other tools

Next steps | Team to make changes integrating our findings. Then, conduct another round of testing with prototypes built in code.

Key Takeaways | severity

Tab manager

Mega menu



Quickly finding a tab



Tab manager helps horizontal tab users find a tab more quickly



Number of participants who made the connection between not pinned and speed: 2



Transitioning to vertical tabs



There is a discoverability problem with vertical tabs, not a desirability problem



Doesn't meet the design principle of minimizing the number of different states



Revisiting a closed tab & workspaces



Number of participants who couldn't discover workspaces: 16



Number of participants who noticed these features require extra clicks: 6

Next steps | Team to make changes integrating our findings. Then, conduct another round of testing with prototypes built in code.

Thank you!

Appendix

APPENDIX 1

More Materials and Participant Details

Test Protocol p1 of 4

We are exploring some new ideas for how your browser could look. Please answer the questions honestly and in as much detail as you can.

- 1.To start, talk about the last time you had a lot of tabs open. What were you doing at the time? [Verbal response]
- 2.Next, open the Edge browser. Show us what you do to organize your tabs when you have a lot open. [Verbal response]
- 3.Do you know of any other tools you could use to organize your tabs? When and why would you use those tools? [Verbal response]
- 4.Next, show us what you do to find a specific tab when there are a lot open. [Verbal response]
- 5.Do you know of any other tools you could use to find a specific tab? When and why would you use those tools? [Verbal response]
- 6.How often do you reopen recently closed tabs in Edge? Talk about the last time you remember doing it. [Verbal response]
- 7.Show us what you would do if you needed to reopen a tab you closed. [Verbal response]
- 8.Do you know of any other tools you could use to reopen a tab you closed? When and why would you use those tools? [Verbal response]
- 9.How familiar are you with the vertical tabs feature in Edge? Is this something you have used before? Why or why not? [Verbal response]
- 10.Show us how you would switch to vertical tabs in your Edge browser. If you haven't done it before, talk about any reactions you have. [Verbal response]
- 11.Think of a time when vertical tabs might help you in the future. What could it help you with and what couldn't it help you with? [Verbal response]
- 12.How familiar are you with the Workspaces feature in Edge? [Verbal response]
- 13.Show us how you would enter a Workspace in your Edge browser. Is this something you have done before? Why or why not? [Verbal response]
- 14.Think of a time when Workspaces might help you in the future. What could it help you with and what couldn't it help you with? [Verbal response]
- 15.Launch URL: <https://www.figma.com/proto/Flzuol34aTR5Cpx99Pc4p4/flow-1?page-id=49%3A4972&node-id=49-5083&node-type=canvas&viewport=628%2C304%2C0.15&t=NS9qWjTyveRlqvmR-1&scaling=scale-down&content-scaling=fixed&starting-point-node-id=49%3A5083>
- 16.You're about to see a concept for something called the Tab Manager. The password is "wide-honey-rare-civil". You can press R to repeat the task if you get stuck. You can press Z to change the size of the window. Keep in mind this is a prototype. It may be slow to load and has limited functionality. After the link has loaded, click to go to the next task.
- 17.You're looking for a tab with the title 'How To Install a New Landscape Bed'. Without clicking anything, talk about ways you would try to find this tab using the tools available to you. [Verbal response]

Test Protocol p2 of 4

18. Now, click to show us how you would open How To Install a New Landscape Bed using the tools available. Once you've opened the tab, go to the next task.
[Verbal response]
19. Talk about anything that surprised you about finding the tab. [Verbal response]
20. Considering how you organize your tabs now, what's easier about the Tab Manager? [Verbal response]
21. Considering how you organize your tabs now, what's more difficult about the Tab Manager? [Verbal response]
22. Talk about a situation where this tool could help you find a tab quicker. [Verbal response]
23. Talk about a situation where this tool couldn't help you find a tab quicker. [Verbal response]
24. Is there something about this tool that could be improved to help you find a tab quicker? [Verbal response]
25. What would you expect to see if you typed something into Search your tabs? What kinds of items would appear in the search? [Verbal response]
26. Launch URL: <https://www.figma.com/proto/60RvzhzLqhZOrP8TztLale/flow-2?page-id=0%3A1&node-id=2-18345&node-type=canvas&viewport=566%2C181%2C0.25&t=1pZ7LNEDGLLtkUDB-1&scaling=scale-down&content-scaling=fixed&starting-point-node-id=2%3A18345>
27. Open this link to perform the next set of tasks. The password is "food-simple-vapor-goat" Remember is a prototype. It may be slow to load and has limited functionality.
28. Imagine you were looking for a tab you closed yesterday about DIY garden edging. Without clicking anything, talk about how you would find it again with the tools available to you. [Verbal response]
29. Now, show us how you would open the closed tab about DIY garden edging. Once you're done, go to the next task. [Verbal response]
30. Talk about anything that surprised you about reopening a closed tab. [Verbal response]
31. Considering how you reopen a closed tab now, what's more efficient about the Tab Manager? [Verbal response]
32. Considering how you reopen a closed tab now, what's less efficient about the Tab Manager? [Verbal response]
33. Here you're seeing your tabs organized vertically and horizontally at the same time: Is there something this could help you do or understand? How does or doesn't it help? [Verbal response]
34. Find the three horizontal dots at the top left in the prototype. Click on the dots. Once you've clicked it, go to the next task. [Verbal response]

Test Protocol p3 of 4

35. Can you think of a situation where you would flip to vertical tabs only? What's the situation? [Verbal response]
36. Think about reasons why you don't use vertical tabs right now. Would this change or stay the same if you had this Tab Manager? Why or why not? [Verbal response]
37. Launch URL: <https://www.figma.com/proto/oQQDbLNw39RQeFfjPJHl9z/Flow-3-hints?page-id=0%3A1&node-id=1-18345&node-type=canvas&viewport=407%2C361%2C0.07&t=8gkXH7KdU8phdxai-1&scaling=contain&content-scaling=fixed&starting-point-node-id=1%3A18345>
38. Open this link to perform the next set of tasks. The password is "agent-write-pluck-sale" Remember this is a prototype. It may be slow to load and has limited functionality.
39. Without clicking anything, talk about how you would open a Workspace called "Trip to Japan". [Verbal response]
40. Afterwards, show us how you would open the Workspace called "Trip to Japan". Once you're done, click "Next Task". [Verbal response]
41. Talk about anything that surprised you about opening the workspace. [Verbal response]
42. Think about reasons why you don't use workspaces right now. Would this change or stay the same if you had this tool available? Why or why not? [Verbal response]
43. Find the three horizontal dots next to the Japan Trip label. What do you think you'd see if you clicked there? What would be in the menu that opened? [Verbal response]
44. Next, show us how you would close Tab Manager. Once you're done, go to the next task. [Verbal response]
45. In what situation would you like Tab Manager to be pinned open? Why? [Verbal response]
46. In what situation would like Tab Manager to automatically close? Why? [Verbal response]
47. Imagine that Tab Manager was available to you today. When is the next time you open it? What would the situation be? [Verbal response]
48. Launch URL: <https://www.figma.com/proto/KFTyv12Y5JzioINkMSZEel/flow-4?page-id=0%3A1&node-id=2-58843&node-type=canvas&viewport=1346%2C619%2C0.22&t=vprbRVhEzVPtwD7m-1&scaling=scale-down&content-scaling=fixed&starting-point-node-id=2%3A58780&hotspot-hints=0>

Test Protocol p4 of 4

49. This is a different version of a Tab Manager. Click to open the next concept. The password is “skit-body-window-byte”. Keep in mind this is a prototype. It may be slow to load and has limited functionality.
50. Show us how you would open this version of a Tab Manager. When it’s open, go to the next task. [Verbal response]
51. Talk about anything that surprised you about opening the Tab Manager. [Verbal response]
52. Without clicking anything, talk about how this Tab Manager compares to the one you just saw. [Verbal response]
53. Which Tab Manager can help you manage tab clutter more efficiently: this one or the previous one? Explain your thinking. [Verbal response]
54. Which Tab Manager can help you find a tab more quickly: this one or the previous one? Explain your thinking. [Verbal response]
55. Which Tab Manager can help you recover a closed tab more easily: this one or the previous one? Explain your thinking. [Verbal response]
56. Compared to the one you just saw, is this Tab Manager easier or more difficult to use with workspaces? Explain your thinking. [Verbal response]
57. The next set of questions are only about the second version of the Tab Manager. Considering how you organize your tabs right now, what’s easier about this Tab Manager? [Verbal response]
58. Considering how you organize your tabs now, what’s more difficult about this Tab Manager? [Verbal response]
59. Considering how you open a closed tab now, what’s more efficient about this Tab Manager? [Verbal response]
60. Considering how you open a closed tab now, what’s less efficient about this Tab Manager? [Verbal response]
61. Imagine that this Tab Manager was available to you today. When is the next time you open it? What would the situation be? [Verbal response]
62. Think about reasons why you don’t use vertical tabs right now. Would this change if you had this tool available? Why or why not? [Verbal response]

Screenener

Which browser is your primary and default internet browser on your computer? [Edge]
Which of the following features have you used in the past month on Edge? [non-tab non-workspace users CANNOT choose tab grouping, workspaces, vertical tabs, auto clean, edge saver, tab search.]
How many tabs do you usually have open on one Edge browser window at a time?

Participant demographics: workspace & tab group users

Participant number and username	P1: Aldi	P2: Chess	P3: JamaicaCumming1507	P4: revkb	P5: keim44	P6: Emtec
Gender	Female	Male	Female	Male	Male	Male
Age	31	30	37	34	34	23
Household income	\$80,000 - \$99,999	\$80,000 - \$99,999	\$40,000 - \$59,999	\$100,000 - \$124,999	\$100,000 - \$124,999	\$20,000 - \$39,999
Employment status	Employed full-time (30 or more hours per week)	Employed full-time (30 or more hours per week)	Employed full-time (30 or more hours per week)	Employed full-time (30 or more hours per week)	Employed full-time (30 or more hours per week)	Employed full-time (30 or more hours per week)
Job function	Finance	Consulting	Sales	Education	Engineering	Sales
Seniority	Manager	Senior	Manager	Chief X officer (CxO)	Director	Manager
Company size	5001-10,000 employees	501-1000 employees	1001-5000 employees	501-1000 employees	201-500 employees	51-200 employees
Industry	Chemicals	Management Consulting	Financial Services	Education Management	Education Management	Consumer Services
Country	Germany	United Kingdom	United Kingdom	United States	United States	Brazil
State				VA	CA	
Social networks	Facebook, LinkedIn, Pinterest	Facebook, LinkedIn	Chrome, Firefox, Internet explorer, Edge	Facebook, Twitter	Facebook, Twitter, LinkedIn, Pinterest	Facebook, Twitter
Web browsers	Chrome, Firefox, Edge	Chrome, Edge		Chrome, Edge	Chrome, Firefox, Edge	Chrome, Firefox, Safari, Opera, Edge
Gaming genres	Puzzles, Ville	Arcade, Casino, Puzzles, Simulation		Arcade, Casino, Puzzles, Simulation, Ville	Arcade, Casino, Puzzles, Simulation, Ville	Arcade, Puzzles, Simulation
Languages spoken	English, Spanish, German, Italian	English, Italian	English	English	English	English, Portuguese
Children	Male, 10 years old			Female, 7 years old	Male, 13 years old; Female, 9 years old	
Devices	Windows, Android Phone, Android Tablet, Smart TV (with built-in apps), Streaming TV box (Roku, Apple TV, etc.)	Windows, iPhone, iPad, Streaming TV box (Roku, Apple TV, etc.)	iPhone, Windows, Smart TV (with built-in apps)	Windows, Android Phone, iPad, Handheld Game Console, Home Game Console, Smart TV (with built-in apps)	Mac, Windows, iPhone, Handheld Game Console, Home Game Console, Smart TV (with built-in apps), Streaming TV box (Roku, Apple TV, etc.)	Android Tablet, Home Game Console, Smart TV (with built-in apps), Windows, iPhone, Streaming TV box (Roku, Apple TV, etc.)

(part 1 of 3)

Participant demographics: non workspace & tab group users

Participant number and username	P7: rama0999	P8: jooliejules47	P9: AnasZar	P10: pokattel	P11: jancave77	P12: sfmke1990
Gender	Male	Female	Male	Female	Female	Male
Age	38	50	35	46	47	34
Household income	\$125,000 - \$149,999	\$60,000 - \$79,999	Less than \$19,999	\$60,000 - \$79,999	\$200,000 +	\$40,000 - \$59,999
Employment status	Employed full-time (30 or more hours per week)	Employed full-time (30 or more hours per week)	Not employed, looking for work	Employed full-time (30 or more hours per week)	Homemaker/stay-at-home parent	Employed full-time (30 or more hours per week)
Job function	Marketing	Education	Business development	Finance	Marketing	Support
Seniority	Manager	Senior	Entry-level	Senior	Manager	Entry-level
Company size	501-1000 employees	201-500 employees	51-200 employees	1001-5000 employees	201-500 employees	10,001+ employees
Industry	Marketing and Advertising	E-Learning	Management Consulting	Financial Services	Marketing and Advertising	Insurance
Country	Australia	United Kingdom	Sweden	United States	Australia	United States
State				CO		WI
Social networks	Facebook, Twitter, LinkedIn	Facebook, Twitter, LinkedIn, Pinterest	Facebook, LinkedIn, Pinterest	Facebook, Twitter, LinkedIn	Facebook, Pinterest	Facebook, Twitter, LinkedIn
Web browsers	Chrome, Edge	Chrome, Firefox, Safari, Opera, Edge	Chrome, Firefox, Safari, Edge	Chrome, Edge	Chrome, Safari, Edge	Chrome, Firefox, Safari, Edge
Gaming genres	Simulation	Puzzles, Simulation	Puzzles, Simulation	Puzzles	Casino	Arcade, Puzzles, Simulation
Languages spoken	English	English	English, Arabic, Swedish	English, Hindi	English	English
Children	Male, 8 years old	Female, 14 years old; Female, 12 years old		Female, 17 years old	Male, 13 years old	Male, 12 years old
Devices	Windows, Android Phone, Home Game Console, Smart TV (with built-in apps)	Windows, iPhone, iPad, Android Phone	Windows, Android Phone, Android Tablet, Handheld Game Console, Home Game Console, Smart TV (with built-in apps), Streaming TV box (Roku, Apple TV, etc.)	Windows, Android Phone, Smart TV (with built-in apps), iPad, Streaming TV box (Roku, Apple TV, etc.)	Windows, iPhone, Android Tablet, Smart TV (with built-in apps)	Windows, Android Phone, Home Game Console, Smart TV (with built-in apps)

(part 2 of 3)

Participant demographics: non workspace & tab group users

Participant number and username	P13: yamsyamsyams	P14: Colorado_Mountains	P15: Calhoon	P16: arakis
Gender	Male	Male	Male	Male
Age	36	23	57	24
Household income	Less than \$19,999	\$175,000 - \$199,999	\$80,000 - \$99,999	\$20,000 - \$39,999
Employment status	Full time student	Full time student	Employed full-time (30 or more hours per week)	Employed part-time (Under 30 hours per week)
Job function			Information technology	Arts and design
Seniority			Entry-level	Entry-level
Company size	any	any	5001-10,000 employees	11-50 employees
Industry			Mechanical or Industrial Engineering	Graphic Design
Country	Canada	United States	United States	India
State		CO	WI	
Social networks	Twitter, Linkedin			Facebook, Twitter, Linkedin, Pinterest
Web browsers	Firefox, Edge	Chrome, Safari, Edge	Chrome, Firefox, Edge	Chrome, Firefox, Safari, Opera, Edge
Gaming genres	Arcade, Casino, Puzzles, Simulation, Ville	Arcade, Puzzles, Simulation	Puzzles, Simulation	Arcade, Casino, Puzzles, Simulation
Languages spoken	English	English	English	English
Children				
Devices	Mac, Windows, Android Phone, Handheld Game Console	Windows, iPhone, iPad, Home Game Console	Windows, Android Phone	Windows, Android Phone, Smart TV (with built-in apps)

(part 3 of 3)

APPENDIX 2

Extra Slides

User Priorities

View

Tabs as a list, switch to vertical tabs and back

Find & Organize

Search, Tidy up, Save current window as workspace

Manage & Switch

Tabs, windows and workspaces, revisit tabs in recently closed & other devices

Awareness

Of these features

Participants

How they manage tab clutter, in their own words:

10

Non-workspaces & non-tab group users

“When I'm really busy and I've got a lot of tabs open, the length of the tab title shortens and I can no longer see, I gather them up and put them into a document, take a break and start again” –P8

“The usual way I implement [organize] is keeping the most important one least on left” –P9

“I don't do anything, I'll have tabs upon tabs, will be hopping between them” –P15

6

Workspaces & tab groups users

“I use the workspaces option, sometimes also the action menu” –P1

“I would try to group everything together and click on the group I need, then expand the group” –P2

“I use the organize and search tab buttons in the top left” –P5

Participants

Here is what our user groups are doing right now:

	Managing tab clutter	Tab search	Revisiting a closed tab
Non-workspaces & non-tab group users	Favorites Organizing from left to right Close some tabs off	Hover preview Memorizing the tabs Opening each tab Alt tab	History Favorites
Workspaces & tab groups users	Favorites Workspaces Tab grouping	Hover preview Memorizing the tabs Tab search Tab grouping	History Favorites

Reactions to tab manager

Both user groups called out ways the tab manager was easier than what they do now, and ways it may fall short:



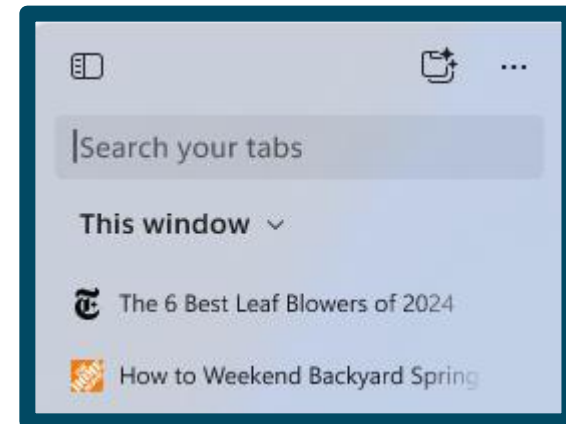
Looks neat

2 participants felt it looks polished and professional



Screen real estate

2 participants observed that depending on how it's pinned it may consume too much screen real estate



Ways tab manager is easier or more difficult than how they manage tab clutter now

Both user groups called out ways the tab manager was easier than what they do now, and ways it may fall short:



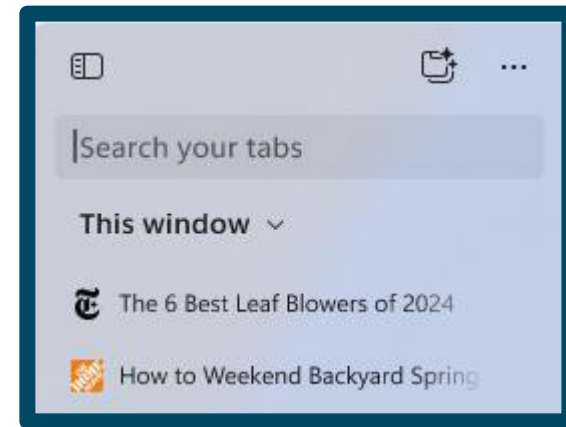
Seeing more of the tab names

8 participants observed they could see more of the tab names when they are laid out vertically



Not different enough from vertical tabs

1 participant remarked it was similar to vertical tabs right now, and many others used similar terms and phrases to describe the current state of vertical tabs and tab manager



Insight | Tab manager solves the problem of tab clutter in a similar that vertical tabs does

On the first try

6 participants (less than half) knew where to look for the tool on their first try

Horizontal tabs

No surprise that participants tried to click through the horizontal tabs

Right click a tab

Some participants tried to right click a horizontal tab to find menu options there

Three dot menu

Where they're used to seeing search history and other options

Vertical tabs

Non-users of vertical tabs talked about how they would try to turn it on

Task
Find tab manager

Other icons

Many participants clicked a few icons in the toolbar before tab manager

Recommendation | Enhance the discoverability of the tab manager entry point, especially for non-tab group users

Non-workspace users won't change habits

Non-workspace users talked about whether they would change their habits with the new tab manager:

Awareness

4 participants mentioned they might use it now that they're aware of this feature

Irrelevant

2 participants felt that this feature was for business use only and wouldn't solve any of their needs

Inefficient

1 participant felt this buried tabs in another part of their browser, when they already had trouble with visibility

"I have to get to a whole other group and search again, essentially I'm doing 2 searches now"

- P13

Insight | If we can enhance the findability of workspaces, tab manager can help tab heavy users discover this feature.